# Solar Na appendix

Sun · Na · downloadable appendix

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|---|---|
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| **Subject** | Solar photospheric abundance appendix · Na |
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Solar sodium: 6.312 ± 0.038

Asplund, Amarsi & Grevesse 2021: 6.22 ± 0.03 · this measurement +0.092 dex

The three-dimensional non-LTE products apply published line-by-line corrections (Asplund et al. 2021; Canocchi et al. 2026) to our one-dimensional fits; they are not our own three-dimensional calculation. The H- and K-band products sit about 0.4 dex below the visible ones. They are shown as measured, with their full error bars.

Measured the way Asplund, Amarsi & Grevesse (2021) measure it: the Asplund et al. 2021 line set (the five Na I lines of their Table A.1), 5 lines measured on one spectrum, each part weighted by its uncertainty.

Na I: 6.291 ± 0.044 · Kitt Peak (1984, molecfit) · VIS · Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

Na I: 6.328 ± 0.040 · Kitt Peak (1984, molecfit) · NIR · Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

## Highlighted Products

### near-UV

**6.671 ± 0.140**

Na I · Deep-graded lines · 1D NLTE (Gerber) · Kitt Peak (Kurucz 2005) · Deep Grade

### VIS

**6.287 ± 0.044**

Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

### red-optical

**6.338 ± 0.088**

Na I · Codex-graded lines · 1D NLTE (Gerber) · Kitt Peak (1984, molecfit) · Codex Grade

### NIR

**6.328 ± 0.040**

Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

### H

**5.779 ± 0.693**

Na I · Codex-graded lines · 1D NLTE (Amarsi) · CRIRES+ H · Codex Grade

### K

**5.903 ± 0.135**

Na I · Codex-graded lines · 1D NLTE (Amarsi) · CRIRES+ · Codex Grade

## Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

| Grade | What it means | This element |
|---|---|---|
| Reference Grade | lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used | 30 products -- Na I · Asplund et al. 2021 lines |
| Codex Grade | lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep | 54 products -- Na I · Codex-graded lines |
| Deep Grade | the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere | 28 products -- Na I · Deep-graded lines |

## Error-bar Forest Plot

### near-UV — Kitt Peak (Kurucz 2005)
### A(Na) dex; axis 6.14–6.89

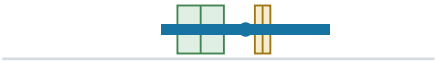
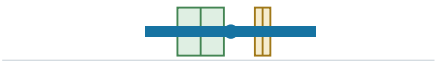
| | | |
|---|---|---|
| Na I · Deep-graded lines · 1D LTE<br>Deep Grade · 2 lines |  | 6.689<br>±0.153 |
| Na I · Deep-graded lines · 1D NLTE (Amarsi)<br>Deep Grade · 2 lines |  | 6.574<br>±0.160 |
| Na I · Deep-graded lines · 1D NLTE (Gerber)<br>Deep Grade · 2 lines |  | 6.671<br>±0.140 |
| Na I · Deep-graded lines · 1D LTE (Gerber)<br>Deep Grade · 2 lines |  | 6.713<br>±0.143 |

### near-UV — Kitt Peak (1984, molecfit)
### A(Na) dex; axis 6.14–6.89

| | | |
|---|---|---|
| Na I · Deep-graded lines · 1D LTE<br>Deep Grade · 2 lines |  | 6.465<br>±0.173 |
| Na I · Deep-graded lines · 1D NLTE (Amarsi)<br>Deep Grade · 2 lines |  | 6.350<br>±0.179 |
| Na I · Deep-graded lines · 1D NLTE (Gerber)<br>Deep Grade · 2 lines |  | 6.437<br>±0.161 |
| Na I · Deep-graded lines · 1D LTE (Gerber)<br>Deep Grade · 2 lines |  | 6.489<br>±0.160 |

near-UV axis: 6.14 · 6.33 · 6.51 · 6.70 · 6.89 A(Na) dex

## VIS — HARPS
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 6.278<br>±0.109 |
| Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 6.264<br>±0.111 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 6.226<br>±0.112 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 4 lines |  | 6.259<br>±0.110 |
| Na I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 4 lines |  | 6.282<br>±0.109 |
| Na I · Deep-graded lines · 1D LTE<br>Deep Grade · 2 lines |  | 6.228<br>±0.036 |
| Na I · Deep-graded lines · 3D NLTE (Amarsi)<br>Deep Grade · 2 lines |  | 6.239<br>±0.103 |
| Na I · Deep-graded lines · 1D NLTE (Amarsi)<br>Deep Grade · 1 line |  | 6.102<br>±0.073 |
| Na I · Deep-graded lines · 1D NLTE (Gerber)<br>Deep Grade · 2 lines |  | 6.218<br>±0.033 |
| Na I · Deep-graded lines · 1D LTE (Gerber)<br>Deep Grade · 2 lines |  | 6.243<br>±0.033 |
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 22 lines |  | 6.370<br>±0.080 |
| Na I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 6.264<br>±0.081 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 22 lines |  | 6.295<br>±0.082 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 20 lines |  | 6.320<br>±0.072 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 22 lines |  | 6.380<br>±0.102 |

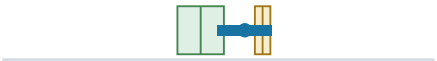
## VIS — IAG FTS
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 1 line |  | 6.394<br>±0.040 |
| Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 1 line |  | 6.386<br>±0.042 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 1 line |  | 6.362<br>±0.040 |

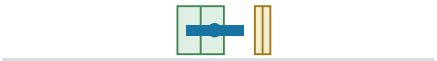
## VIS — IAG FTS
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 1 line |  | 6.391<br>±0.042 |
| Na I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 1 line |  | 6.401<br>±0.040 |
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 10 lines |  | 6.354<br>±0.138 |
| Na I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 1 line |  | 6.386<br>±0.042 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 10 lines |  | 6.318<br>±0.139 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 10 lines |  | 6.347<br>±0.129 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 10 lines |  | 6.353<br>±0.130 |

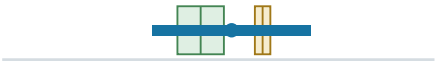
## VIS — IAG FTS
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 3 lines |  | 6.293<br>±0.038 |
| Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 6.281<br>±0.041 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 6.246<br>±0.034 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 3 lines |  | 6.277<br>±0.035 |
| Na I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 3 lines |  | 6.298<br>±0.037 |
| Na I · Deep-graded lines · 1D LTE<br>Deep Grade · 2 lines |  | 6.191<br>±0.036 |
| Na I · Deep-graded lines · 3D NLTE (Amarsi)<br>Deep Grade · 2 lines |  | 6.202<br>±0.103 |
| Na I · Deep-graded lines · 1D NLTE (Amarsi)<br>Deep Grade · 1 line |  | 6.066<br>±0.072 |
| Na I · Deep-graded lines · 1D NLTE (Gerber)<br>Deep Grade · 2 lines |  | 6.139<br>±0.030 |
| Na I · Deep-graded lines · 1D LTE (Gerber)<br>Deep Grade · 2 lines |  | 6.207<br>±0.033 |
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 6.288<br>±0.040 |

## VIS — IAG FTS
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 5 lines |  | 6.281<br>±0.049 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 6.238<br>±0.037 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 6 lines |  | 6.272<br>±0.038 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 6 lines |  | 6.292<br>±0.040 |

## VIS — Kitt Peak (Kurucz 2005)
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 6.300<br>±0.042 |
| Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 6.287<br>±0.044 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 6.248<br>±0.043 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 4 lines |  | 6.284<br>±0.042 |
| Na I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 4 lines |  | 6.305<br>±0.041 |
| Na I · Deep-graded lines · 1D LTE<br>Deep Grade · 2 lines |  | 6.250<br>±0.033 |
| Na I · Deep-graded lines · 3D NLTE (Amarsi)<br>Deep Grade · 2 lines |  | 6.260<br>±0.103 |
| Na I · Deep-graded lines · 1D NLTE (Amarsi)<br>Deep Grade · 1 line |  | 6.104<br>±0.073 |
| Na I · Deep-graded lines · 1D NLTE (Gerber)<br>Deep Grade · 2 lines |  | 6.236<br>±0.034 |
| Na I · Deep-graded lines · 1D LTE (Gerber)<br>Deep Grade · 2 lines |  | 6.263<br>±0.031 |
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 18 lines |  | 6.300<br>±0.081 |
| Na I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 6.287<br>±0.048 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 18 lines |  | 6.248<br>±0.083 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 18 lines |  | 6.284<br>±0.080 |

## VIS — Kitt Peak (Kurucz 2005)
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 18 lines |  | 6.305<br>±0.080 |

## VIS — Kitt Peak (1984, molecfit)
## A(Na) dex; axis 5.96–6.52

| | | |
|---|---|---|
| Na I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 6.304<br>±0.042 |
| Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 6.291<br>±0.044 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 6.252<br>±0.043 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 4 lines |  | 6.278<br>±0.043 |
| Na I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 4 lines |  | 6.309<br>±0.042 |
| Na I · Deep-graded lines · 1D LTE<br>Deep Grade · 2 lines |  | 6.221<br>±0.036 |
| Na I · Deep-graded lines · 3D NLTE (Amarsi)<br>Deep Grade · 2 lines |  | 6.232<br>±0.103 |
| Na I · Deep-graded lines · 1D NLTE (Amarsi)<br>Deep Grade · 1 line |  | 6.095<br>±0.072 |
| Na I · Deep-graded lines · 1D NLTE (Gerber)<br>Deep Grade · 2 lines |  | 6.210<br>±0.031 |
| Na I · Deep-graded lines · 1D LTE (Gerber)<br>Deep Grade · 2 lines |  | 6.235<br>±0.033 |
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 17 lines |  | 6.294<br>±0.086 |
| Na I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 6.291<br>±0.047 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 17 lines |  | 6.247<br>±0.088 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 16 lines |  | 6.263<br>±0.071 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 17 lines |  | 6.298<br>±0.081 |

VIS axis: 5.96 · 6.10 · 6.24 · 6.38 · 6.52 A(Na) dex

## red-optical — IAG FTS
## A(Na) dex; axis 5.70–6.54

| | | |
|---|---|---|
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 4 lines |  | 6.327<br>±0.163 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 5.990<br>±0.261 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 4 lines |  | 6.332<br>±0.173 |

## red-optical — Kitt Peak (Kurucz 2005)
## A(Na) dex; axis 5.70–6.54

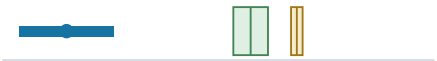
| | | |
|---|---|---|
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 4 lines |  | 6.322<br>±0.054 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 6.231<br>±0.037 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 4 lines |  | 6.271<br>±0.079 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 4 lines |  | 6.327<br>±0.054 |

## red-optical — Kitt Peak (1984, molecfit)
## A(Na) dex; axis 5.70–6.54

| | | |
|---|---|---|
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 5 lines |  | 6.364<br>±0.063 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 3 lines |  | 6.262<br>±0.073 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 6.338<br>±0.088 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 5 lines |  | 6.370<br>±0.062 |

red-optical axis: 5.70 · 5.91 · 6.12 · 6.33 · 6.54 A(Na) dex

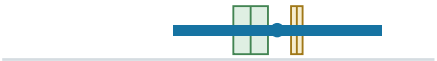
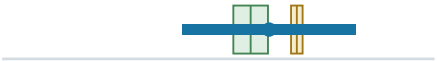
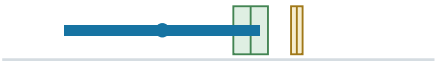
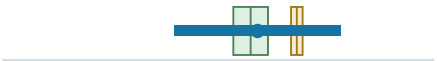
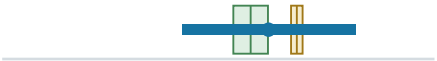
## NIR — CRIRES+ J
## A(Na) dex; axis 5.79–6.54

| | | |
|---|---|---|
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 6.213<br>±0.294 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 1 line |  | 5.901<br>±0.081 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 2 lines |  | 6.196<br>±0.281 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 2 lines |  | 6.215<br>±0.294 |

## NIR — CRIRES+ Y (full arm)
## A(Na) dex; axis 5.79–6.54

| | | |
|---|---|---|
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 1 line |  | 6.123<br>±0.041 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 1 line |  | 6.105<br>±0.041 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 1 line |  | 6.108<br>±0.049 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 1 line |  | 6.122<br>±0.041 |

## NIR — Kitt Peak (1984, molecfit)
## A(Na) dex; axis 5.79–6.54

| | | |
|---|---|---|
| Na I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 1 line |  | 6.301<br>±0.036 |
| Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 1 line |  | 6.328<br>±0.040 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 1 line |  | 6.295<br>±0.036 |
| Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 1 line |  | 6.266<br>±0.180 |
| Na I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 1 line |  | 6.303<br>±0.036 |
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines |  | 6.252<br>±0.150 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 6.067<br>±0.170 |
| Na I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 3 lines |  | 6.232<br>±0.145 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines |  | 6.251<br>±0.150 |

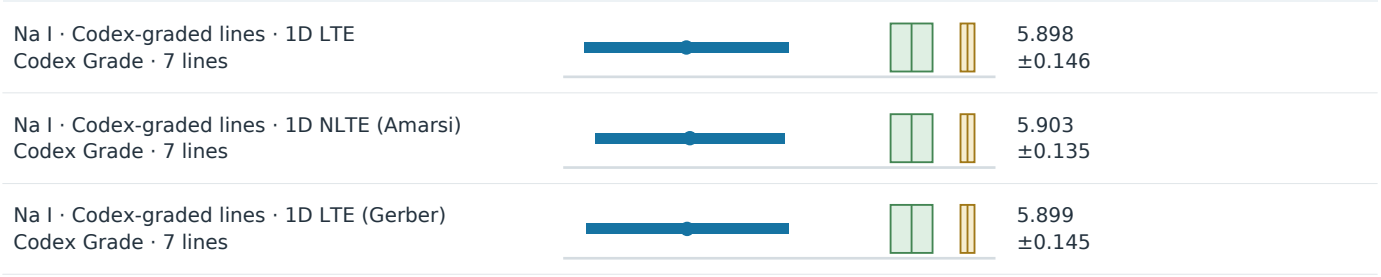
NIR axis: 5.79 · 5.98 · 6.16 · 6.35 · 6.54 A(Na) dex

## H — CRIRES+ H
## A(Na) dex; axis 5.06–6.55

| | | |
|---|---|---|
| Na I · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines |  | 5.796<br>±0.692 |
| Na I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 3 lines |  | 5.779<br>±0.693 |
| Na I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 3 lines |  | 5.803<br>±0.713 |

H axis: 5.06 · 5.43 · 5.80 · 6.17 · 6.55 A(Na) dex

**K — CRIRES+**
**A(Na) dex; axis 5.72–6.34**




K axis: 5.72 · 5.88 · 6.03 · 6.19 · 6.34 A(Na) dex

**How to read this forest.** Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

## How this was measured

Every sodium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the sodium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

| Band | What was measured | Solar spectra | Treatments | Lines | Result A(Na) | Largest uncertainty terms (dex) |
|---|---|---|---|---|---|---|
| near-UV | Na I · Deep-graded lines | Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE | 2 | 6.350–6.671 (1D-NLTE) | fit / line-to-line scatter 0.14, microturbulence 0.08 |
| VIS | Na I · Deep-graded lines | HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE, 3D-NLTE | 1–2 | 6.202–6.260 (3D-NLTE) | fit window 0.32, model atmosphere 0.08 |
| VIS | Na I · Codex-graded lines | HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE, 3D-NLTE | 1–22 | 6.264–6.386 (3D-NLTE) | continuum placement 0.03, fit / line-to-line scatter 0.03 |

| Band | What was measured | Solar spectra | Treatments | Lines | Result A(Na) | Largest uncertainty terms (dex) |
|---|---|---|---|---|---|---|
| VIS | Na I · Asplund et al. 2021 lines | HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE, 3D-NLTE | 1–4 | 6.264–6.386 (3D-NLTE) | continuum placement 0.04, NLTE / 3D correction 0.03 |
| red-optical | Na I · Codex-graded lines | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE | 2–5 | 5.990–6.338 (1D-NLTE) | continuum placement 0.22, fit window 0.20 |
| NIR | Na I · Codex-graded lines | CRIRES+ J, CRIRES+ Y (full arm), Kitt Peak (1984, molecfit) | 1D-LTE, 1D-NLTE | 1–3 | 5.901–6.232 (1D-NLTE) | fit / line-to-line scatter 0.14, fit window 0.07 |
| NIR | Na I · Asplund et al. 2021 lines | Kitt Peak (1984, molecfit) | 1D-LTE, 1D-NLTE, 3D-NLTE | 1 | 6.328 (3D-NLTE) | NLTE / 3D correction 0.03, fit / line-to-line scatter 0.02 |
| H | Na I · Codex-graded lines | CRIRES+ H | 1D-LTE, 1D-NLTE | 3 | 5.779 (1D-NLTE) | fit / line-to-line scatter 0.69, telluric residual 0.54 |
| K | Na I · Codex-graded lines | CRIRES+ | 1D-LTE, 1D-NLTE | 7 | 5.903 (1D-NLTE) | fit / line-to-line scatter 0.13, telluric residual 0.12 |

## Solar sodium context

Sodium is traced by neutral Na I lines from the near-ultraviolet to the K band. The reference lines are the five Asplund et al. 2021 used (4751, 5148, 6154, 6160 and 10746 Å), with the oscillator strengths they adopted. Na I departs from LTE, so each spectrum also carries non-LTE products; each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

## Download the evidence

Download PDF — Solar Na appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

## Reproducibility

Generated from the pinned science snapshot. Source commit d00d21cb0a6fa21d09513efd1520256ce66dc7d7
Feed Na.json v1.131 · updated 2026-10-27T15:01:19Z
Generated 2026-10-28T16:38:16-06:00 (source commit timestamp)
Feed SHA-256 4380d74f7bd9af7e2e727f6ffca1e2982139120427e3854a0d93cd756b0166aa
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

## References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

**Solar abundance and indicator analyses.** Solar Na comparison literature; not an adopted Codex abundance. Sources: [1], [2].

**Atomic data and gf authorities.** Na I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

**Model and grid sources.** Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

**Model and grid sources.** Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Na_Amarsi2020_PySME.csv). Sources: [5].

**Model and grid sources.** Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

**Model and grid sources.** Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

**Solar abundance and indicator analyses.** Published line set NA_AGSS21 (the line selection). Sources: [1].

**Model and grid sources.** Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte, A_1d_lte Sources: [1], [6].

**Model and grid sources.** Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte Sources: [1].

Product-to-source index

H · CRIRES+ H · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-fb210d1eee02

H · CRIRES+ H · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-a80b76db5d6a

H · CRIRES+ H · Na I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-c047829fd0f2

K · CRIRES+ · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-0552befecb14

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red-optical · Kitt Peak (1984, molecfit) · Na I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-1582fe7c0f11

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**[1]** — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

**[2]** — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

## Atomic data and gf authorities

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## Model and grid sources

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